

Table 7: Ablation study on the backbone architecture of student models. We run experiments three times with different random seeds and report the average of F1 and AUPRC scores. For results including standard deviations, please refer to Table 11 in Appendix D.

Model	Total	Backbone	Embedding	OAI		ToxicChat		HarmBench		WildGuardMix		Average	
				F1	AUPRC	F1	AUPRC	F1	AUPRC	F1	AUPRC	F1	AUPRC
DeBERTa-large + HarmAug	435M	304M	131M	0.7236	0.8791	0.6283	0.7553	0.8331	0.8841	0.7576	0.8265	0.7357	0.8362
DeBERTa-xsmall + HarmAug	71M	22M	49M	0.6475	0.8102	0.4322	0.6270	0.7947	0.8378	0.7025	0.7600	0.6442	0.7588
DeBERTa-small + HarmAug	142M	44M	98M	0.6782	0.8459	0.5349	0.6996	0.8025	0.8484	0.6971	0.7863	0.6782	0.7950
DeBERTa-base + HarmAug	184M	86M	98M	0.7066	0.8485	0.5776	0.7112	0.8160	0.8690	0.7368	0.8089	0.7093	0.8094
BERT-base + HarmAug	110M	86M	24M	0.6442	0.7837	0.5081	0.6353	0.7891	0.8480	0.6985	0.7735	0.6600	0.7601
BERT-large + HarmAug	335M	303M	32M	0.6606	0.8074	0.5532	0.6702	0.8118	0.8587	0.7171	0.7975	0.6857	0.7835
RoBERTa-base + HarmAug	125M	86M	39M	0.6726	0.8368	0.5348	0.7022	0.8011	0.8471	0.7383	0.8069	0.6867	0.7983
RoBERTa-large + HarmAug	355M	303M	52M	0.6975	0.8590	0.5428	0.7115	0.8332	0.8715	0.7416	0.8218	0.7038	0.8160
Qwen2-Instruct + HarmAug	494M	358M	136M	0.6940	0.7256	0.5659	0.5523	0.7989	0.8339	0.7054	0.7138	0.6910	0.7064

formance than smaller models, at the cost of increased FLOPs, latency, and peak memory usage, as shown in Table 6. However, this increased cost remains negligible compared to the cost of the teacher model (Llama-Guard-3), with DeBERTa-large demonstrating significantly greater efficiency.

Backbones of student models. We study the effect of different backbone architectures in the student safety guard model q_ϕ . To compare with the DeBERTa-large model used in the main experiments, we also train BERT (Devlin et al., 2019), RoBERTa (Liu et al., 2019), and Qwen2-Instruct (Yang et al., 2024) on both the training dataset $\mathcal{D}$ and our synthetic dataset $\hat{\mathcal{D}}$. As shown in Table 7, DeBERTa-large outperforms both RoBERTa-large and BERT-large across all benchmark datasets, with the exception of HarmBench, where its F1 score is comparable to that of RoBERTa-large. Even the DeBERTa-base model shows a higher F1 and AUPRC scores than BERT-large, with performance comparable to RoBERTa-large. Despite having the largest model size, the Qwen model underperforms compared to the bidirectional encoder models (RoBERTa, and DeBERTa). These experimental results support our choice of DeBERTa as the backbone for the main experiments.

Size of synthetic dataset. In Fig. 7, we plot the average AUPRC across four benchmark datasets (OAI, ToxicChat, HarmBench, and WildGuardMix) while varying the size of the synthetic dataset $\hat{\mathcal{D}}$ from 20,000 to 100,000 examples, sampled from p_{LLM} . The average AUPRC improves as we train the model with more synthetic data, achieving the highest AUPRC with 100,000 synthetic samples. However, the performance gains diminish as the size of synthetic dataset grows. This may be attributed to some redundancy in the synthetic dataset. Improving the diversity of the synthetic dataset by prompting the LLM to generate new samples conditioned on previously generated instances represents a promising direction for future research.

Soft labels. In this experiment, we adjust the temperature of the logits from the teacher p_θ , where logits refer to the pre-softmax values, and perform knowledge distillation using Eq. (1). Increasing the temperature leads to a smoother probability distribution over the output classes of the teacher model. As shown in Fig. 8, a temperature of 0.0, which corresponds to hard labels, shows the best performance compared to other temperature values. Thus, we adopt a hard-labeling strategy for all our experiments.

5 CONCLUSION

In this work, we proposed to distill a large safety guard model into a smaller version for efficient deployment in low resource environments such as mobile devices. To bridge the performance gap between small and large models, we proposed a simple yet effective data augmentation method called HarmAug that involves jailbreaking an LLM and prompting the LLM to generate harmful instructions. In our experiments, the 435M-parameter model trained with HarmAug yielded significant improvements in FLOPs, latency, and GPU memory usage, while maintaining AUPRC and F1 scores comparable to larger models with over 7 billion parameters. Furthermore, the use of our smaller model reduced the runtime of the red-teaming process and enabled more efficient further fine-tuning to defend against new jailbreak attacks.

Figure 7: Average AUPRC across synthetic dataset sizes.

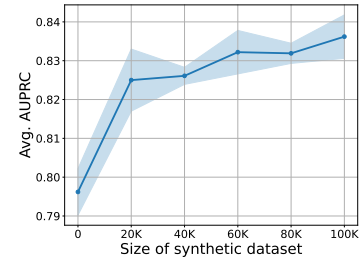
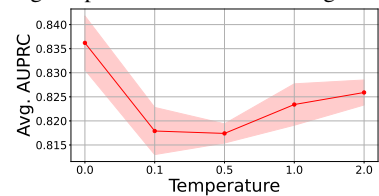


Figure 8: Average AUPRC with varying temperature of the teacher logits.



REPRODUCIBILITY STATEMENT

We use PyTorch (Paszke et al., 2019) and the Transformers library from Hugging Face (Wolf et al., 2020) to implement our proposed method and all the baselines in our experiments. All implementation details are described in the experimental setup part of Sec. 4.1, Sec. 4.2, and Sec. 4.3. We provide anonymous URLs to our [code](#), [safety guard model](#), and , allowing the research community to fully access, reproduce, and extend our work on improving detection of harmful conversations and computational efficiency of safety guard models. Detailed instructions for reproducing our knowledge distillation process are provided in our [code](#).

ETHICS STATEMENT

Our work presents a small and efficient safety guard model designed to detect and mitigate harmful user queries, including jailbreak attacks, aimed at compromising the safety of LLMs. This approach is critical for ensuring that LLMs can be deployed safely in real-world applications. By maintaining performance levels comparable to significantly larger models, our lightweight safety guard model addresses the ethical concerns associated with LLM deployment while significantly reducing computational and financial costs. The reduced resource requirements not only make the model more accessible to organizations with limited infrastructure but also minimize the environmental impact of large-scale model deployment.

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